SONALI SHAW

RF Hardware Design Engineer | Radio Unit Architecture & Validation | 5G NR | LTE | O-RAN | 3GPP Conformance
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PROFESSIONAL SUMMARY

RF Engineer with 4 years of end-to-end Radio Unit (RU) design and validation for LTE and 5G NR sub-6GHz systems. Delivered a complete dual-band B1/B3 RU for a German OEM from architecture through production. Conducted triple-band RU feasibility studies covering sub-1GHz, mid-band FDD, and TDD band combinations. Experienced in PAM board schematic design (Driver PA, Coupler, Circulator, Connector, Switches), Xilinx-AMD RFSoC Frequency Planner for Image Rejection Filter selection, component symbol and decal review in Mentor Graphics Expedition, cross-functional collaboration with HW/SW/Mechanical teams, S-parameter and impedance matching ,smith-chart , RF simulation (AWR/ADS/HFSS), PCB layout review, 3GPP conformance testing, and O-RAN RU validation.

CORE TECHNICAL SKILLS

Standards & Bands: 5G NR, LTE, O-RAN, ORAN RU | Sub-6GHz (FDD & TDD) | 3GPP TS 36.104 / 36.141 / 38.104 / 38.141
RU Design: Multi-Band & Dual-Band RU Architecture, Triple-Band Feasibility, PAM Board Design, Link & RF/Power Budget, Noise Figure, IP3, P1dB, Gain Flatness, PA Design & Analysis, Driver PA, Coupler, Circulator, RF Switch Design, Image Rejection Filter Selection, Intra-Band ICH-CH Isolation, DPD Stability
EDA & Schematic: Mentor Graphics Expedition (Schematic Design, Symbol Creation, Decal Review, BOM/EN Management), Cadence AWR, Keysight ADS, ANSYS HFSS, EM Simulation, S-parameter ,smithchart(impedance matching) ,Xilinx-AMD RFSoC Frequency Planner
Lab & Test Tools: Keysight O-RAN Studio, Keysight VXT, Spectrum Analyzer, VNA, Anritsu Signal Generator & Analyzer, Power Meter
TX Testing: Output Power, EVM, ACLR, SEM/OBUE, IMD, OBW, Spurious Emissions, Power Control
RX Testing: Receiver Sensitivity, Dynamic Range, Blocking, Max ADC Input, BLER, Throughput
PCB & Layout: DFM Rules, Controlled Impedance, RF Grounding, Inter-Band Isolation, Via Strategy, Thermal Mgmt, Copper Coin Measurement
Testing Scope: Standalone Component, TRX Board, Radio Conformance, Unit-Level Functional, Room & Thermal, Single & Multi-Carrier
Collaboration: Cross-Functional Teamwork — HW Design, SW, Mechanical, Layout, System & Vendor Teams | Root Cause Analysis & Debug
Self-Learning: Python scripting for RF instrument automation (Keysight VXT via VISA/SCPI) — CW signal generation, 5G NR waveform configuration (personal project, not deployed in production
Report and Documentation: Test report /Test plan/ technical document

PROFESSIONAL EXPERIENCE

RF Engineer | VVDN Technologies, Bhubaneswar | Jun 2022 – Present

- **RU Design Ownership:** Led end-to-end RF design and validation for 5+ LTE/5G NR Radio Unit projects across multiple sub-6GHz bands — all delivered on schedule and within 3GPP conformance requirements
- **German OEM — Dual-Band RU:** Owned complete design of a combined dual-band Radio Unit (B1 + B3) for a German OEM — RF architecture, dual-band PA, inter-band isolation, and shared RF/power budget from prototype bring-up through production sign-off
- **Triple-Band Feasibility:** Conducted triple-band RU feasibility studies covering sub-1GHz, mid-band FDD, and TDD band combinations — inter-band coexistence, isolation analysis, line-up efficiency, combined RF/link/power budget, and PA path architecture for multi-band operation
- **RF Architecture & Budget:** Defined radio architecture; performed link budget, RF/power budget, noise figure, IP3, P1dB, gain flatness, and PA gain/efficiency analysis for single-band, dual-band, and triple-band RU configurations
- **PAM Board Schematic Design:** Designed Power Amplifier Module (PAM) board schematics comprising Driver PA, Coupler, Circulator, RF Connector, and Switches — covering component selection, biasing circuits, and signal path architecture for NXP, Innogration, Ampleon, Wa-Tech, and Macom PA variants in Mentor Graphics Expedition
- **Symbol & Decal Review:** Performed component symbol creation and decal (land pattern) review in Mentor Graphics Expedition for new RF components — verified IPC-compliant footprints, pin mapping, and schematic symbols prior to BOM and Engineering Notice (EN) release
- **BOM & EN Management:** Managed Bill of Materials (BOM) and Engineering Notice (EN) creation including component symbol and decal review for schematic implementation and PCB handoff
- **RFSoC Frequency Planning & Image Rejection Filter Selection:** Used Xilinx-AMD RFSoC Frequency Planner to analyse ADC/DAC sampling frequency configurations and identify harmonic and intermodulation spur locations — determined Image Rejection Filter (IRF) specifications and selected appropriate image rejection filters for the Radio Unit digital front-end signal chain
- **Simulation & Correlation:** Executed RF simulations to validate S-parameter ,impedance matching by using smith chart in Cadence AWR, Keysight ADS, and ANSYS HFSS; correlated simulation results with lab measurements to close design gaps across all frequency bands
- **PCB Layout Review:** Reviewed PCB layouts applying RF and DFM rules — controlled impedance, grounding, inter-band isolation, via design, thermal management, and copper coin measurement for heat dissipation
- **TRX & Thermal Testing:** Performed standalone component testing, RF TRX board testing, and integrated unit-level functional test cases under room and thermal conditions including DPD stability validation
- **3GPP Conformance:** Performed RF conformance testing (single and multi-carrier) per TS 36.104/36.141/38.104/38.141 — TX: Output Power, EVM, ACLR, SEM, IMD, OBW, OBUE, Spurious; RX: Sensitivity, Dynamic Range, Blocking, Max ADC Input, BLER, Throughput
- **O-RAN Validation:** Conducted O-RAN RU testing using Keysight O-RAN Studio — C-plane/U-plane configuration, SCP and PCAP file generation, BLER calculation, and PTP synchronization validation for complete O-RAN RU bring-up
- **Cross-Functional Debug:** Collaborated with HW design, SW, and mechanical teams to perform system-level RF debug and root cause analysis — PA thermal failures, spurious emissions, inter-band interference, and PCB layout anomalies
- **Report and Documentation:** Prepared Radio Conformance and technical test reports ,test plans and maintained test-data sheets to document test configurations, measurements, results, and observations wrt Customer requirements

KEY PROJECTS

Dual-Band B1/B3 5G NR Radio Unit | German OEM | Prototype to Production

End-to-end design of a single RU supporting dual-band operation (B1 + B3) for a German OEM — full RF signal chain including PAM board design (Driver PA, Coupler, Circulator, Switches), RFSoC frequency planning for image rejection filter selection, inter-band isolation strategy, and shared RF/link/power budget from prototype through production.

Dual-band RU architecture | PAM board schematic (Driver PA, Coupler, Circulator, Switches) | RFSoC frequency planning & IRF selection | Symbol & decal review | RF simulation (AWR/HFSS) | PCB layout review | Single & multi-carrier testing | O-RAN bring-up | 3GPP TS 38.104/38.141

Triple-Band RU Feasibility Study | German OEM | Pre-Development Phase

Feasibility study for next-generation triple-band RU configurations covering sub-1GHz, mid-band FDD, and TDD band combinations — inter-band coexistence, isolation, line-up efficiency, and multi-band RF/power budget analysis.

Sub-1GHz | Mid-band FDD | TDD bands | Triple-band RF architecture | Inter-band coexistence & isolation | Multi-band RF/power budget

5G NR and LTE Multi-Band Radio Units | End-to-End RF Ownership

Radio architecture | Link & power budget | PAM board schematic (Driver PA, Coupler, Circulator, Switches) | RFSoC frequency planning & IRF selection | Symbol & decal review in Expedition | RF simulation | PCB layout review | Cross-functional debug (HW/SW/Mechanical) | 3GPP conformance (TS 38.104/38.141 | TS 36.104/36.141)

SELF-LEARNING & DEVELOPMENT

RF Instrument Automation using Python | Personal Project

Developed Python scripts for automated RF instrument control using Keysight VXT via VISA/SCPI interface — CW signal generation at defined frequencies and power levels, and 5G NR waveform configuration. Undertaken independently to build RF automation skills and explore test efficiency improvement techniques.

Skills: Python | VISA / SCPI | Keysight VXT | CW Signal Generation | 5G NR Waveform Control | RF Test Automation

EDUCATION

B.Tech – Electrical & Electronics Engineering (EEE)

International Institute of Information Technology Bhubaneswar (IIIT BBSR) | 2018 – 2022